

MERIDIAN: A Multimodal Evidence-Grounded Framework for Reliable Medical Diagnosis with Uncertainty and Selective Abstention

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Abstract

Multimodal medical foundation models have demonstrated the potential of jointly learning representations from medical imaging, clinical language, and structured patient information, enabling transferable representations for downstream clinical tasks [1], [2], [3], [4]. However, accurate prediction alone does not establish whether a model’s output should be trusted in the presence of ambiguous cases, inconsistent modalities, missing information, or distribution shifts. We present MERIDIAN (Multimodal Evidence-grounded Reliable Intelligent Diagnosis with Integrated Abstention Networks), an independent reliability-oriented framework designed to augment multimodal medical AI with explicit mechanisms for uncertainty estimation, cross-modal evidence consistency, distribution-shift detection, and selective prediction. MERIDIAN jointly processes CT representations, radiology reports, and structured clinical features through modality-specific encoders and gated cross-modal attention, followed by prediction and uncertainty heads. Building on Bayesian uncertainty estimation through Monte Carlo dropout and aleatoric–epistemic uncertainty modeling [5], [6], the framework introduces a Cross-Modal Evidence Alignment (CMEA) signal, conformal prediction for statistically controlled prediction sets [7], [8], distribution-shift detection using Maximum Mean Discrepancy and energy-based representations [9], [10], explicit missing-modality awareness, and a selective prediction mechanism that enables the system to predict, escalate, or abstain when sufficient evidence is unavailable [11], [12]. The framework additionally incorporates structured counterfactual evaluation and an integrated reliability dashboard to expose the evidence and uncertainty associated with each prediction. The current prototype uses simulated CT feature representations and synthetic multimodal samples to validate the proposed reliability architecture rather than claiming clinical deployment readiness. Experimental evaluation investigates prediction performance, uncertainty discrimination, conformal coverage, distribution-shift sensitivity, modality robustness, calibration, and risk–coverage behavior. Results provide initial evidence that explicit reliability modeling can complement multimodal prediction by making uncertainty, evidence consistency, and model limitations more measurable and interpretable. MERIDIAN therefore establishes a reproducible research framework for developing multimodal medical AI systems that can recognize not only what they predict, but also when their predictions may not be sufficiently reliable.

Keywords: Multimodal Medical AI, Medical Vision–Language Models, Evidence-Grounded Diagnosis, Uncertainty Quantification, Cross-Modal Evidence Alignment, Conformal Prediction, Distribution Shift Detection, Selective Prediction, Abstention, Missing Modality Learning, Clinical Decision Support.

1. INTRODUCTION

Multimodal medical artificial intelligence (AI) has emerged as a promising direction for clinical decision support by combining complementary information from medical images, clinical text,

and structured patient data. Rather than relying on a single modality, multimodal systems can learn relationships between visual findings and their corresponding clinical descriptions, potentially producing richer representations

for downstream diagnostic tasks [2], [3], [4], [13]. Recent advances in medical vision-language foundation models have further demonstrated the value of large-scale multimodal pretraining. In particular, Merlin introduced a computed tomography (CT) vision-language foundation model capable of learning transferable representations from paired CT studies and radiology reports, establishing an important direction for multimodal medical representation learning [1]. Despite these advances, strong predictive performance does not necessarily indicate that a model’s output is reliable in every clinical situation.

Medical AI systems may encounter ambiguous examinations, incomplete patient information, disagreement between imaging and clinical descriptions, and data distributions that differ from those observed during training. Conventional evaluation based primarily on metrics such as accuracy or area under the receiver operating characteristic curve (AUC) provides limited information about whether an individual prediction should actually be trusted. This is particularly important in clinical environments, where an incorrect but highly confident prediction may be more problematic than an explicit indication that sufficient evidence is unavailable. Consequently, reliable medical AI requires mechanisms that can quantify uncertainty, identify potentially unreliable inputs, detect distributional changes, and provide an appropriate mechanism for refusing or escalating uncertain cases.

Uncertainty estimation provides one approach to addressing this limitation. Bayesian approximations based on Monte Carlo dropout can estimate predictive uncertainty in neural networks, while subsequent work has distinguished between aleatoric uncertainty arising from inherent data ambiguity and epistemic uncertainty associated with model uncertainty [5], [6]. Deep ensembles provide another effective approach for predictive uncertainty estimation [14]. However, uncertainty estimation alone does not determine whether the different modalities supporting a prediction are mutually consistent. A model may remain confident even when information from an

image and its associated report is contradictory or when one modality contains insufficient evidence.

Cross-modal representation learning provides the foundation for explicitly examining relationships between medical images and language. Contrastive image-text learning has demonstrated that paired medical images and reports can be used to learn aligned representations [2], while frameworks such as GLoRIA and MedCLIP have further explored multimodal representation learning using paired and unpaired medical data [3], [13]. Transformer-based attention mechanisms provide a flexible means of learning interactions between heterogeneous representations [15], [16]. These developments motivate the investigation of whether cross-modal agreement itself can be used as an additional signal of prediction reliability.

Another important challenge is distribution shift. A model trained on one population, institution, acquisition protocol, or data-generation process may encounter substantially different observations during deployment. Statistical approaches such as Maximum Mean Discrepancy (MMD) provide a principled method for comparing distributions in representation spaces [9], while energy-based approaches have been investigated for detecting out-of-distribution examples in neural networks [10], [17]. Explicitly detecting such shifts can provide an additional warning signal when a model is operating outside the conditions under which its predictions were learned.

Clinical data are also frequently incomplete. A patient may have an imaging study but no corresponding report, a report without complete structured clinical variables, or missing information in one or more modalities. Previous research has investigated learning under missing modalities and heterogeneous multimodal inputs [18], [19], [20], [21]. Treating missing information as an ordinary input or silently replacing it with an estimated value can obscure an important source of uncertainty. A reliable multimodal system should therefore explicitly recognize which modalities are available and adapt its representation and reliability estimates

accordingly.

Beyond estimating uncertainty, a medical AI system should be capable of determining when it should not make a definitive prediction. Selective classification introduces the concept of a reject option, allowing a model to abstain from predictions for cases that are considered unreliable [11], [12]. Conformal prediction provides a complementary framework for constructing prediction sets with statistical coverage guarantees under its underlying assumptions [7], [8]. Together, these approaches provide a basis for moving from a conventional “predict” paradigm toward a more cautious decision process in which a system can predict, escalate a case for expert review, or abstain when available evidence is insufficient.

Calibration is similarly important because a probability estimate is useful only when it meaningfully reflects the likelihood of correctness. Modern neural networks can exhibit substantial miscalibration even when their classification performance is strong, motivating explicit evaluation and calibration of predictive probabilities [22]. Metrics such as the Brier score provide additional means of evaluating probabilistic predictions [23]. For a reliability-oriented medical AI system, therefore, performance, calibration, uncertainty, distributional robustness, and selective behavior should be evaluated jointly rather than independently.

Counterfactual evaluation provides another perspective on model reliability by examining how predictions respond to controlled modifications of the input evidence. Counterfactual explanations have been proposed as a means of understanding how changes to input factors can affect model decisions [24], while causal reasoning provides a broader theoretical foundation for distinguishing meaningful relationships from associations [25]. In multimodal medical systems, structured perturbations of image representations, clinical variables, or disease-related textual information can therefore be used to investigate whether predictions respond to clinically relevant evidence rather than solely to superficial correlations. Such experiments should be

interpreted as robustness and sensitivity analyses rather than as proof of causal relationships.

Despite substantial progress in multimodal medical representation learning, uncertainty estimation, out-of-distribution detection, and selective prediction, these components are often studied separately. There remains a need for an integrated framework that treats reliability as a multidimensional property involving evidence consistency, uncertainty, modality availability, distributional similarity, calibration, and the ability to abstain. This motivates **MERIDIAN** (Multimodal Evidence-grounded Reliable Intelligent Diagnosis with Integrated Abstention Networks), an independent research framework designed around this reliability-oriented perspective.

MERIDIAN is inspired by the multimodal medical representation-learning direction demonstrated by Merlin, while addressing a complementary research question: *given a multimodal medical prediction, can the system explicitly determine whether the available evidence is sufficiently reliable to justify that prediction?* The framework combines modality-specific representations for CT, radiology reports, and structured clinical information using gated cross-modal attention. On top of the predictive representation, MERIDIAN introduces a reliability layer incorporating Cross-Modal Evidence Alignment (CMEA), Monte Carlo uncertainty estimation, conformal prediction, distribution-shift detection, missing-modality awareness, selective prediction and abstention, structured counterfactual evaluation, and an integrated reliability dashboard.

The central objective of MERIDIAN is therefore not simply to maximize diagnostic accuracy, but to make the reliability of individual predictions measurable and transparent. The framework distinguishes between cases where the available evidence supports a confident prediction and cases where uncertainty, missing information, cross-modal inconsistency, or distributional shift warrants additional scrutiny. The current implementation is intentionally presented as a research prototype: its CT component uses simulated feature representations rather than

real three-dimensional CT volumes, and its experiments do not constitute clinical validation. Instead, the study establishes a reproducible computational framework for investigating how explicit reliability mechanisms can complement multimodal medical AI and provide a foundation for future evaluation using real-world CT, radiology-report, and clinical datasets such as LIDC-IDRI, NLST, and MIMIC-based resources [26], [27], [28], [29], [30].

2. LITERATURE REVIEW

2.1. Multimodal Medical Representation Learning

The integration of heterogeneous clinical modalities has become an important research direction in medical artificial intelligence, motivated by the complementary information contained in medical images, clinical narratives, and structured patient data. Multimodal machine learning provides a general framework for learning from such heterogeneous sources and has been studied extensively across different application domains [4]. In medical imaging, paired image–text learning has demonstrated that radiological images and their associated reports can provide mutually informative supervision for representation learning. Zhang et al. proposed contrastive learning of medical visual representations from paired images and text, demonstrating that semantic information contained in clinical reports can improve learned visual representations [2]. GLORIA further investigated global and local alignment between medical images and textual descriptions for label-efficient medical image recognition [3]. MedCLIP extended this direction by investigating contrastive learning from unpaired medical images and text, reducing the dependence on strictly paired datasets [13].

Large-scale medical vision–language foundation models have subsequently expanded the scope of multimodal representation learning. Merlin introduced a CT vision–language foundation model and dataset based on large-scale CT studies and radiology reports, demonstrating the potential of multimodal pretraining for transferable CT representations [1].

These developments establish an important foundation for multimodal clinical AI; however, representation quality and downstream predictive performance do not by themselves indicate whether an individual model output is sufficiently reliable. MERIDIAN builds upon this broader multimodal direction while focusing specifically on the reliability of multimodal predictions.

2.2. Vision–Language Fusion and Cross-Modal Attention

Effective fusion of heterogeneous modalities requires mechanisms capable of modeling interactions between their learned representations. Attention-based architectures provide a flexible approach for allowing one representation to selectively incorporate information from another. The Transformer architecture established self-attention as a powerful mechanism for modeling relationships among input representations [15]. Subsequent multimodal architectures extended attention mechanisms to visual and linguistic representations. ViLBERT introduced separate visual and textual streams connected through co-attentional transformer layers, demonstrating the value of cross-modal interactions for vision–language representation learning [31]. The Multimodal Transformer further investigated attention-based fusion of heterogeneous modalities and demonstrated the ability to model interactions across multiple input streams [16].

These developments motivate the gated cross-modal attention mechanism used in MERIDIAN. Rather than simply concatenating independently generated image, text, and structured clinical embeddings, MERIDIAN explicitly models cross-modal interactions and learns modality-dependent gates. This design also provides a mechanism for representing modality availability and examining how information from different sources contributes to the final prediction.

2.3. Medical Imaging and Clinical Language Encoders

The effectiveness of multimodal systems depends strongly on the quality of their

modality-specific encoders. Transformer-based language models such as BERT established bidirectional contextual representation learning for natural language understanding [32]. DistilBERT subsequently provided a smaller and computationally efficient distilled variant suitable for resource-constrained applications [33]. In the clinical domain, ClinicalBERT investigated adapting contextual language representations to clinical notes and downstream hospital prediction tasks [34], while RadBERT specifically investigated transformer-based language modeling for radiological text [35]. These models demonstrate the importance of domain-specific language representations for medical applications.

For three-dimensional medical imaging, pretrained models such as Med3D have investigated transfer learning for 3D medical image analysis [36], while Swin Transformer-based self-supervised pretraining has provided another approach for learning transferable representations from volumetric medical images [37]. More broadly, deep learning has become extensively established in medical image analysis, encompassing classification, detection, segmentation, and representation learning [38]. In the current MERIDIAN prototype, a lightweight language encoder and simulated CT representations are used to validate the proposed reliability architecture. Real volumetric encoders such as Med3D or 3D Swin-based architectures represent a natural future implementation rather than a component claimed to have been evaluated in the current prototype.

2.4. Uncertainty Estimation in Deep Learning

Predictive uncertainty is an important component of reliable machine learning because a model’s confidence may not accurately reflect the probability that its prediction is correct. Gal and Ghahramani introduced Monte Carlo dropout as a practical Bayesian approximation for estimating uncertainty in deep neural networks [5]. Kendall and Gal further distinguished between aleatoric uncertainty, associated with inherent ambiguity in observations, and epistemic uncertainty,

associated with uncertainty in the model itself [6]. Deep ensembles provide another scalable approach to predictive uncertainty estimation and have demonstrated strong empirical performance across a range of tasks [14].

MERIDIAN incorporates Monte Carlo dropout to obtain multiple stochastic predictions and separates predictive uncertainty into aleatoric and epistemic components. This allows the framework to distinguish cases that are intrinsically ambiguous from cases for which the model has comparatively limited knowledge. The resulting uncertainty estimates are subsequently combined with other reliability signals rather than being treated as the sole criterion for determining whether a prediction should be accepted.

2.5. Calibration and Probabilistic Reliability

A model can achieve strong classification performance while producing poorly calibrated confidence estimates. Guo et al. demonstrated that modern neural networks can be substantially miscalibrated and investigated methods for improving the relationship between predicted confidence and empirical correctness [22]. The Brier score provides a complementary probabilistic evaluation measure by assessing the accuracy of probabilistic forecasts rather than only their final class decisions [23]. Calibration is particularly important for clinical decision-support systems because confidence values may influence whether predictions are accepted, reviewed, or rejected.

Accordingly, MERIDIAN evaluates predictive reliability in addition to conventional classification performance. Calibration analysis is integrated with uncertainty estimation and selective prediction so that a high-confidence output is not automatically treated as reliable without evaluating whether the confidence estimates themselves are meaningful.

2.6. Conformal Prediction

Conformal prediction provides a distribution-free framework for constructing prediction sets with formal coverage properties under its exchangeability assumptions. The foundations of algorithmic conformal prediction were

established by Vovk et al. [8], while more recent work has provided accessible formulations and theoretical perspectives on conformal prediction and its coverage guarantees [7]. Instead of forcing a model to produce a single class when evidence is ambiguous, conformal methods can return a set containing multiple plausible outcomes.

MERIDIAN incorporates conformal prediction as a complementary safety mechanism. A calibrated prediction set is generated for each case, allowing the framework to represent diagnostic ambiguity explicitly. The conformal component is therefore combined with uncertainty, cross-modal consistency, modality availability, and distribution-shift signals to support the final reliability decision. The resulting coverage statement is interpreted under the statistical assumptions of conformal prediction and is not presented as a clinical guarantee.

2.7. Distribution Shift and Out-of-Distribution Detection

Medical AI systems may encounter data distributions that differ from the training environment because of changes in patient populations, imaging protocols, institutions, equipment, or clinical workflows. Detecting such distribution shifts is therefore important for assessing whether a model is operating within a familiar representation space. Maximum Mean Discrepancy (MMD) provides a kernel-based statistical approach for comparing probability distributions [9]. Energy-based statistical approaches provide another family of distributional comparison methods based on distances between observations [39]. In the context of neural networks, energy scores have also been investigated for detecting out-of-distribution examples [10], while Hendrycks and Gimpel established an early baseline for detecting misclassified and out-of-distribution examples using predictive behavior [17].

MERIDIAN uses MMD and energy-based measurements in its fused representation space to investigate whether a new case differs substantially from the training distribution. Rather than assuming that a prediction is

reliable solely because the classifier produces a high probability, the framework treats detected distributional deviation as an additional warning signal.

2.8. Learning with Missing Modalities

Complete multimodal information is not always available in clinical environments. Missing imaging, incomplete reports, or unavailable structured clinical variables can substantially alter the information available to a predictive model. Previous research has explored learning with heterogeneous or missing modalities in medical applications. Havaei et al. investigated hetero-modal image segmentation, demonstrating the importance of handling varying modality configurations [18]. ModDrop introduced modality dropout as a mechanism for improving robustness to missing modalities during multimodal learning [19]. More recent medical studies have examined multimodal learning with missing data for cancer diagnosis [20], while SimMLM investigated multimodal learning under missing-modality conditions [21].

MERIDIAN addresses this issue through explicit modality gates. The system records which modalities are available, adapts the fusion process accordingly, and incorporates modality availability into its reliability assessment. This design aims to prevent missing information from being silently interpreted as evidence and instead makes incomplete multimodal input an explicit component of the reliability analysis.

2.9. Selective Prediction and Abstention

Traditional classification assumes that a model should produce a prediction for every input. Selective classification relaxes this assumption by allowing a model to reject or abstain from predictions for cases where its estimated reliability is insufficient. Geifman and El-Yaniv investigated selective classification for deep neural networks and established risk-coverage analysis as a means of evaluating the relationship between prediction risk and the fraction of cases on which the model chooses to make predictions [11]. SelectiveNet subsequently incorporated an integrated reject option directly into a neural network architecture [12].

This concept is central to MERIDIAN. Instead of treating prediction as a binary process of either correct or incorrect classification, MERIDIAN introduces three operational states: *PREDICT*, *ESCALATE*, and *ABSTAIN*. These decisions are informed by multiple reliability signals, including predictive uncertainty, cross-modal evidence alignment, modality availability, and conformal prediction-set characteristics. Risk-coverage analysis is subsequently used to evaluate whether restricting predictions to higher-reliability cases reduces the observed prediction risk.

2.10. Counterfactual and Evidence-Sensitivity Analysis

Understanding whether a medical AI system responds to clinically meaningful evidence is an important aspect of trustworthy model evaluation. Counterfactual explanation research has proposed modifying input factors to investigate how automated decisions change under alternative conditions [24]. Causal modeling provides a broader theoretical framework for reasoning about relationships between variables and interventions [25]. However, counterfactual sensitivity analysis in a predictive model should not automatically be interpreted as evidence of causality.

MERIDIAN therefore employs structured counterfactual perturbations as an evaluation mechanism rather than claiming causal identification. Examples include removing disease-related textual information, perturbing CT-derived representations, and modifying structured clinical variables. The purpose is to determine whether model predictions respond appropriately to changes in relevant evidence and remain comparatively stable under changes that should not substantially affect the diagnostic interpretation.

2.11. Clinical Datasets and Future Validation

Several publicly available datasets provide potential foundations for future real-world evaluation of multimodal medical AI. LIDC-IDRI provides a reference database containing thoracic CT examinations and annotated pulmonary nodules [26]. The National

Lung Screening Trial established a large-scale context for evaluating low-dose CT screening and demonstrated the clinical significance of CT-based lung cancer detection [27], [28]. The LUNA16 challenge subsequently provided a standardized benchmark for pulmonary nodule detection algorithms [40]. MIMIC-IV provides a large de-identified electronic health record resource [29], while MIMIC-CXR combines chest radiographs with free-text radiology reports [30].

These datasets illustrate possible directions for extending MERIDIAN from simulated representations toward real multimodal clinical data. However, because the current prototype does not claim to have been trained and clinically validated on these datasets, they are considered future validation resources rather than evidence for the current experimental results.

2.12. Research Gap and Positioning of MERIDIAN

The literature demonstrates substantial progress in multimodal medical representation learning, vision-language modeling, uncertainty estimation, calibration, distribution-shift detection, missing-modality learning, conformal prediction, and selective classification. However, these research directions are commonly investigated as separate components. Multimodal foundation models such as Merlin primarily address the problem of learning powerful transferable representations from large-scale medical image-text data [1], whereas uncertainty and selective prediction methods address different aspects of predictive reliability [5], [7], [11]. Similarly, distribution-shift detection and missing-modality learning provide complementary safeguards but do not independently provide a unified reliability decision mechanism.

MERIDIAN is positioned at the intersection of these directions. Its contribution is not a replacement for multimodal foundation models, nor does it claim to reproduce Merlin. Instead, it investigates an independent reliability layer that combines cross-modal evidence alignment, uncertainty decomposition, conformal prediction, distribution-shift

detection, missing-modality awareness, selective abstention, and structured counterfactual evaluation. By integrating these signals into a common decision framework, MERIDIAN aims to investigate whether multimodal medical AI can move beyond the question of *what the model predicts* toward the equally important question of *whether the prediction should be trusted*.

3. METHODOLOGY AND DATA

3.1. Overall Framework

MERIDIAN is designed as a multimodal reliability-oriented medical AI framework that integrates imaging, clinical language, and structured clinical information with an explicit reliability layer. The complete computational pipeline consists of data generation and preparation, modality-specific representation learning, gated cross-modal fusion, multimodal prediction, uncertainty estimation, evidence alignment, distribution-shift analysis, missing-modality handling, conformal prediction, selective abstention, calibration, counterfactual evaluation, and ablation analysis.

The current prototype is intentionally designed as a reproducible research framework. Since the present implementation does not use a large-scale real-world CT foundation model, CT information is represented using simulated feature vectors. This allows the reliability architecture and its individual components to be evaluated independently before future extension to real three-dimensional CT volumes and clinical datasets.

The complete architecture and experimental workflow are summarized in Fig. 1.

3.2. Multimodal Dataset Construction

MERIDIAN operates on three complementary information sources: CT-derived representations, radiology reports, and structured electronic health record (EHR) features. Each modality is processed independently before multimodal fusion. The CT component represents imaging information through a simulated feature representation in the current prototype, while the text component represents clinical language extracted from radiology reports. Structured

clinical variables provide additional patient-level information.

The dataset construction process generates multimodal samples containing the three modality representations together with their diagnostic labels. To prevent the current prototype from being interpreted as a real clinical deployment system, all experimental conclusions are restricted to the simulated-data setting. Public datasets such as LIDC-IDRI, NLST, MIMIC-IV, and MIMIC-CXR are considered potential resources for future real-world validation [26], [27], [29], [30].

3.3. Modality-Specific Representation Learning

3.3.1 CT Representation

The imaging branch represents each CT examination using a fixed-dimensional feature vector. In a future real-data implementation, this component can be replaced with a pretrained three-dimensional medical image encoder such as Med3D or a 3D Swin Transformer-based architecture [36], [37]. The current representation is used to evaluate the reliability architecture without claiming that the prototype performs direct volumetric CT analysis.

3.3.2 Radiology Text Representation

Radiology reports are converted into contextual text representations using a transformer-based language encoder. The current implementation uses DistilBERT because of its comparatively lightweight architecture [33], while domain-specific alternatives such as ClinicalBERT and RadBERT provide possible future extensions for clinical and radiological language modeling [34], [35].

3.3.3 Structured Clinical Representation

Structured clinical variables are transformed into a compact numerical representation using a dedicated EHR encoder. This branch allows demographic and clinical information to contribute independently to the final multimodal representation while maintaining explicit separation from image and text information.

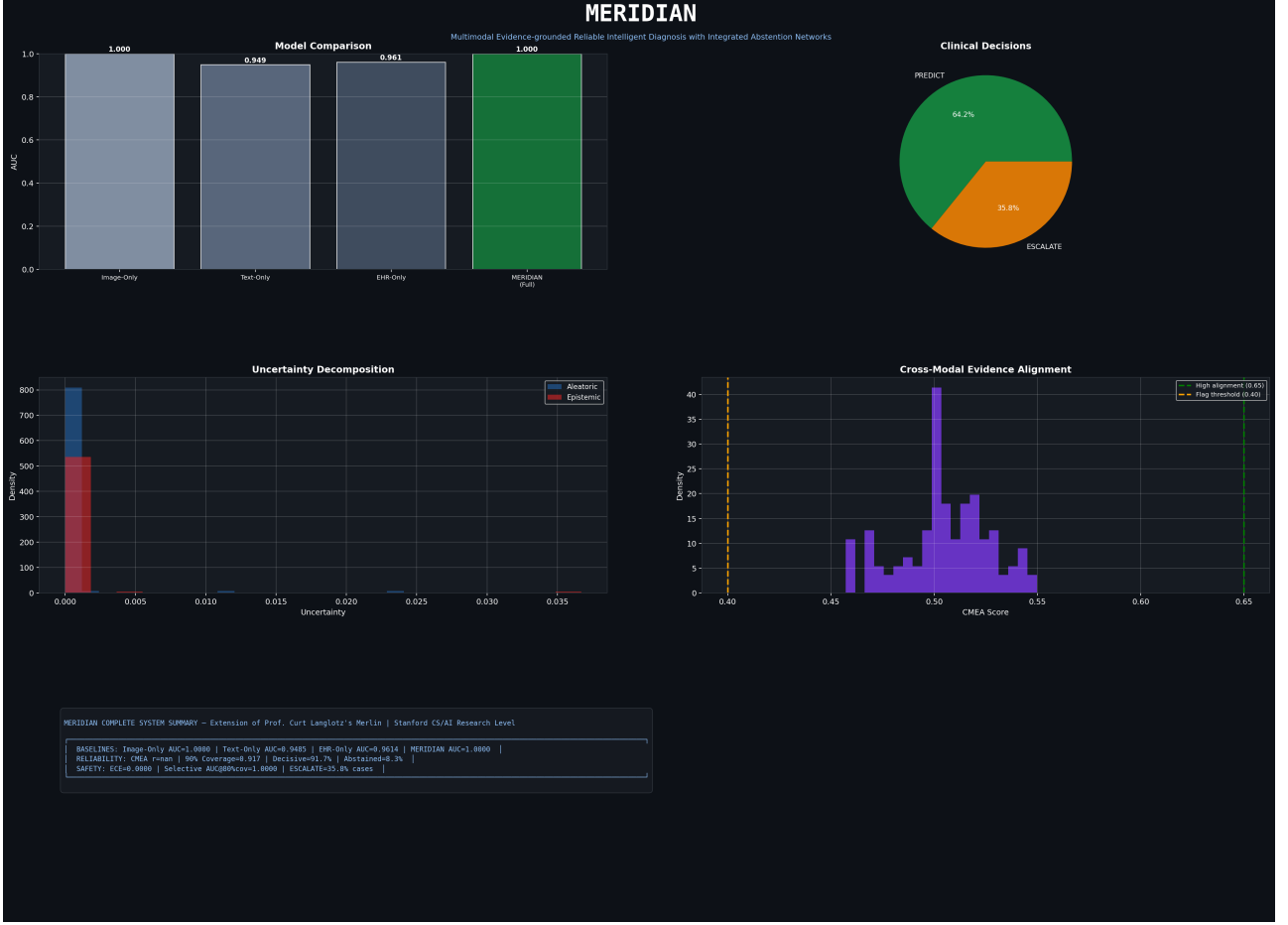


Figure 1. Overall MERIDIAN framework and experimental workflow, illustrating multimodal input processing, representation learning, gated cross-modal fusion, prediction, uncertainty estimation, reliability analysis, and the final predict–escalate–abstain decision process.

3.4. Gated Cross-Modal Fusion

The three modality-specific embeddings are integrated using a gated cross-modal attention mechanism. Attention-based architectures provide a general mechanism for learning interactions between heterogeneous representations [15]. Previous vision–language architectures have demonstrated the usefulness of cross-modal attention for jointly representing visual and textual information [16], [31].

In MERIDIAN, the image and text representations interact through cross-modal attention, while the structured clinical representation is incorporated through an additional gated pathway. The resulting fused representation is passed to the prediction and reliability modules. The learned gates additionally provide information about the relative availability and contribution of each modality.

3.5. Multimodal Prediction and Training

The fused representation is processed by a prediction head responsible for the primary diagnostic task. In parallel, an uncertainty head estimates predictive uncertainty, while an auxiliary binary head provides an additional abnormality-oriented prediction signal. The model is trained using a multi-task objective combining the relevant prediction losses and the uncertainty-related objective.

The training process and associated learning curves are shown in Fig. 2.

3.6. Cross-Modal Evidence Alignment

A central component of MERIDIAN is Cross-Modal Evidence Alignment (CMEA), which evaluates the consistency between the learned image and text representations. Given normalized image and text embeddings, their cosine similarity is computed as an

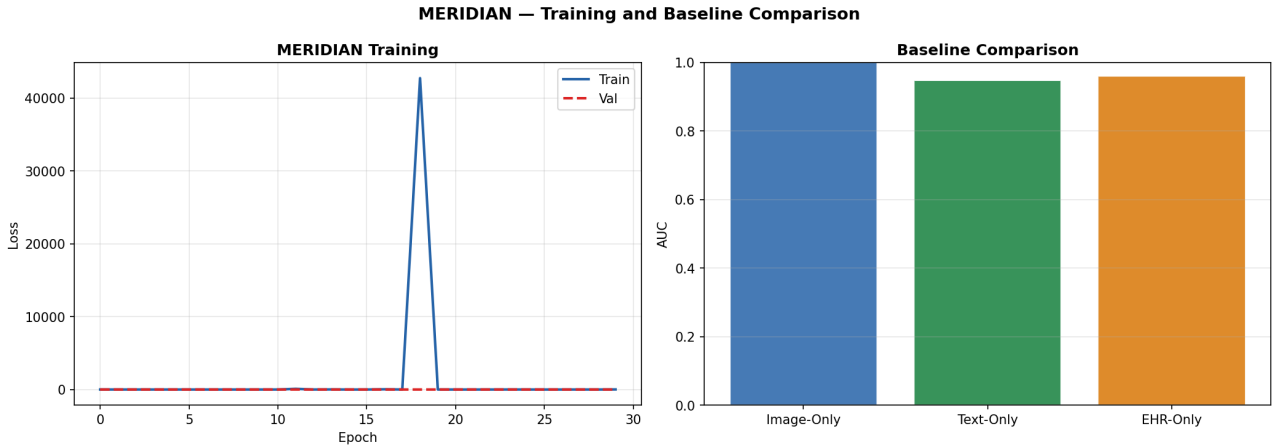


Figure 2. MERIDIAN training behavior, showing the optimization process and model performance across the training procedure.

evidence-alignment score. Higher similarity indicates stronger representational agreement, whereas lower similarity identifies cases in which the information contained in the two modalities may be less consistent.

The CMEA analysis is illustrated in Fig. 3.

CMEA is treated as an additional reliability signal rather than as an independent diagnostic decision. In the current prototype, it is used to investigate whether cross-modal consistency can contribute useful information for reliability assessment.

3.7. Predictive Uncertainty Estimation

MERIDIAN estimates predictive uncertainty using Monte Carlo dropout. Multiple stochastic forward passes are performed while dropout remains active, producing a distribution of predictions rather than a single deterministic output. Monte Carlo dropout provides a practical approximation to Bayesian inference in deep neural networks [5].

The resulting uncertainty is interpreted in terms of epistemic and aleatoric components. Epistemic uncertainty represents uncertainty associated with the model and its limited knowledge, whereas aleatoric uncertainty reflects ambiguity inherent in the observations [6]. Deep ensembles provide an alternative uncertainty-estimation strategy and are considered a possible future comparison [14].

3.8. Calibration Analysis

Prediction probabilities are evaluated for calibration to determine whether model

confidence corresponds appropriately to empirical correctness. Modern neural networks can produce overconfident or poorly calibrated predictions even when their classification performance is strong [22]. MERIDIAN therefore evaluates probabilistic reliability alongside classification performance.

The calibration behavior of the multimodal model is shown in Fig. 4.

3.9. Conformal Prediction

MERIDIAN incorporates conformal prediction to construct prediction sets rather than requiring a single class prediction for every case. Conformal prediction provides statistical coverage guarantees under its underlying assumptions [7], [8].

A calibration subset is used to determine the nonconformity threshold, after which the resulting prediction set is generated for previously unseen samples. At the selected confidence level, the method evaluates whether the true class is contained within the resulting prediction set.

The conformal prediction results are presented in Fig. 5.

The conformal results are interpreted according to the assumptions of the method and are not presented as a clinical safety guarantee.

3.10. Distribution Shift Detection

MERIDIAN evaluates whether incoming samples differ from the training distribution in the learned multimodal representation space.

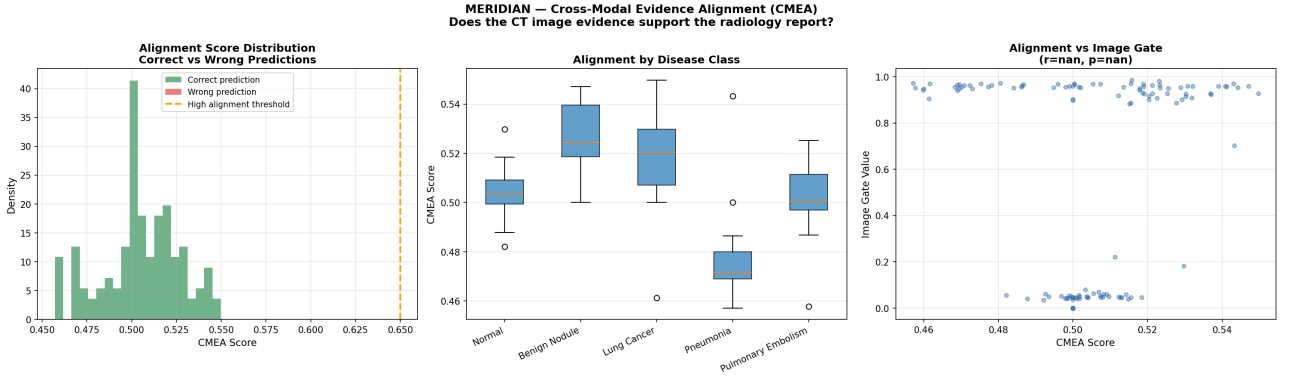


Figure 3. Cross-Modal Evidence Alignment (CMEA) analysis, showing the relationship between image–text representation consistency and the reliability of multimodal predictions.

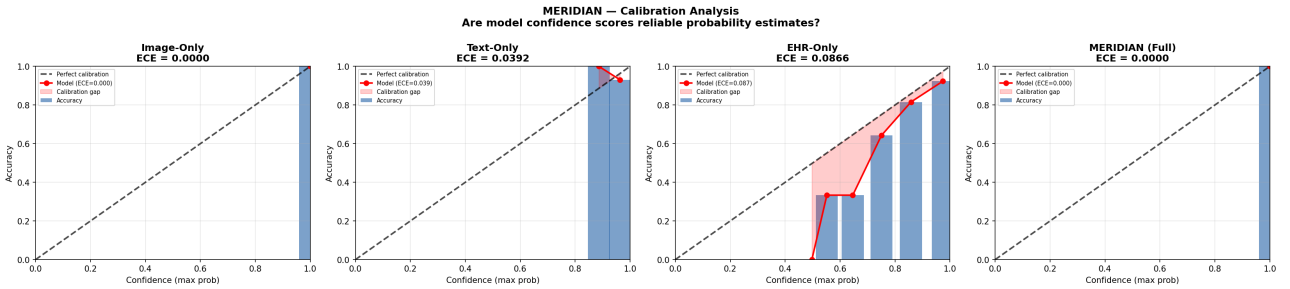


Figure 4. Calibration analysis of MERIDIAN predictions, illustrating the relationship between predicted confidence and observed empirical correctness.

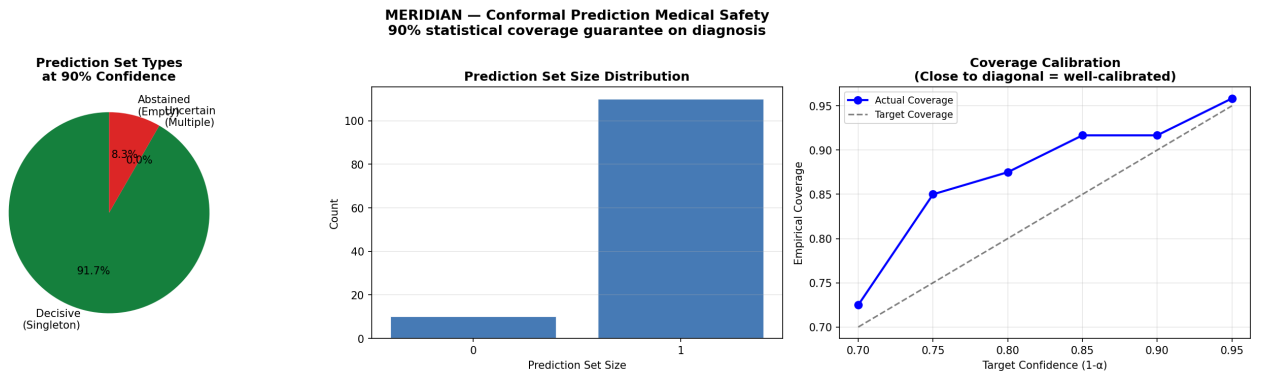


Figure 5. Conformal prediction analysis showing prediction-set behavior and empirical coverage at the selected confidence level.

Maximum Mean Discrepancy (MMD) is used as a statistical measure for comparing distributions [9]. Energy-based distributional analysis is additionally incorporated as a complementary signal, motivated by energy-based approaches to out-of-distribution detection [10], [17].

The resulting distribution-shift analysis is shown in Fig. 6.

A detected distribution shift is treated as a warning signal indicating that model predictions may require additional scrutiny.

3.11. Missing Modality Awareness

Clinical multimodal datasets frequently contain incomplete observations. MERIDIAN explicitly records modality availability and adapts the fusion process according to the available information. Rather than silently replacing unavailable information, the framework incorporates modality availability into the reliability assessment.

Previous studies have investigated robustness to missing or heterogeneous modalities using approaches such as hetero-modal learning and

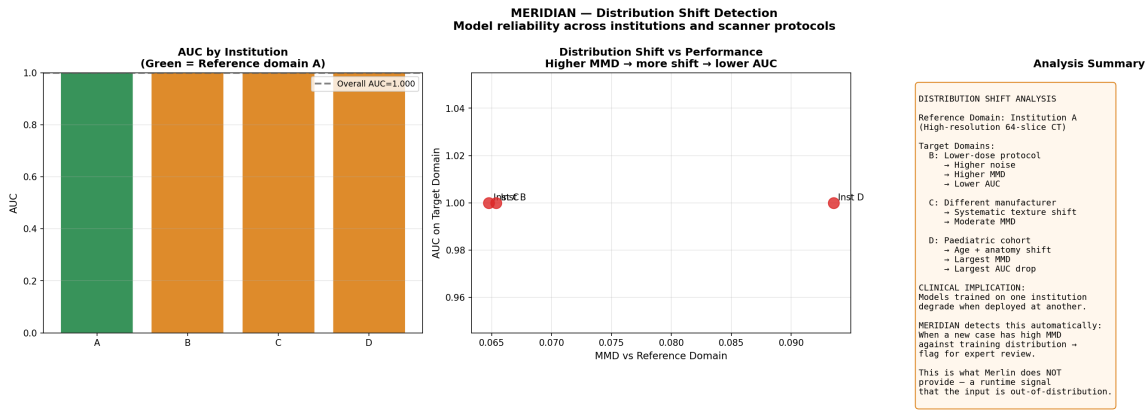


Figure 6. Distribution-shift analysis in the multimodal representation space using statistical and energy-based measures to identify deviations from the training distribution.

modality dropout [18], [19]. More recent work has explored missing-modality learning in medical applications [20], [21].

The missing-modality experiments are summarized in Fig. 7.

3.12. Selective Prediction and Abstention

MERIDIAN combines multiple reliability signals to determine whether a prediction should be accepted. Selective classification allows a model to reject uncertain cases rather than producing a prediction for every input [11]. SelectiveNet further demonstrates the use of an integrated reject option within neural architectures [12].

The MERIDIAN reliability layer considers predictive uncertainty, CMEA, modality availability, and conformal prediction-set characteristics. These signals are combined into a reliability assessment that places each case into one of three states:

1. **PREDICT**: sufficient evidence is available for a high-reliability prediction.
2. **ESCALATE**: the case contains borderline reliability signals and should receive additional expert review.
3. **ABSTAIN**: the available evidence is insufficient to justify a definitive prediction.

The selective prediction and risk-coverage behavior are shown in Fig. 8.

3.13. Structured Counterfactual Evaluation

MERIDIAN performs structured perturbation experiments to investigate the sensitivity of

predictions to different sources of evidence. Textual information can be modified by removing disease-related terms, image representations can be perturbed or suppressed, and structured clinical variables can be modified toward reference values.

Counterfactual explanations provide a general framework for investigating how model decisions change under controlled input modifications [24]. Causal inference provides a broader theoretical basis for intervention-based reasoning [25]; however, the experiments in MERIDIAN are interpreted as structured sensitivity analyses rather than causal identification.

The resulting counterfactual experiments are presented in Fig. 9.

3.14. Ablation Study

An ablation study is conducted to investigate the contribution of the individual reliability components. Components are systematically removed or disabled and the resulting changes in model behavior are compared with the complete MERIDIAN configuration.

The ablation experiments provide a mechanism for determining whether the observed reliability behavior depends on the combined reliability layer or can be attributed to individual components.

The ablation results are shown in Fig. 10.

3.15. Integrated Reliability Evaluation

The outputs of the individual experiments are integrated into a unified reliability assessment.

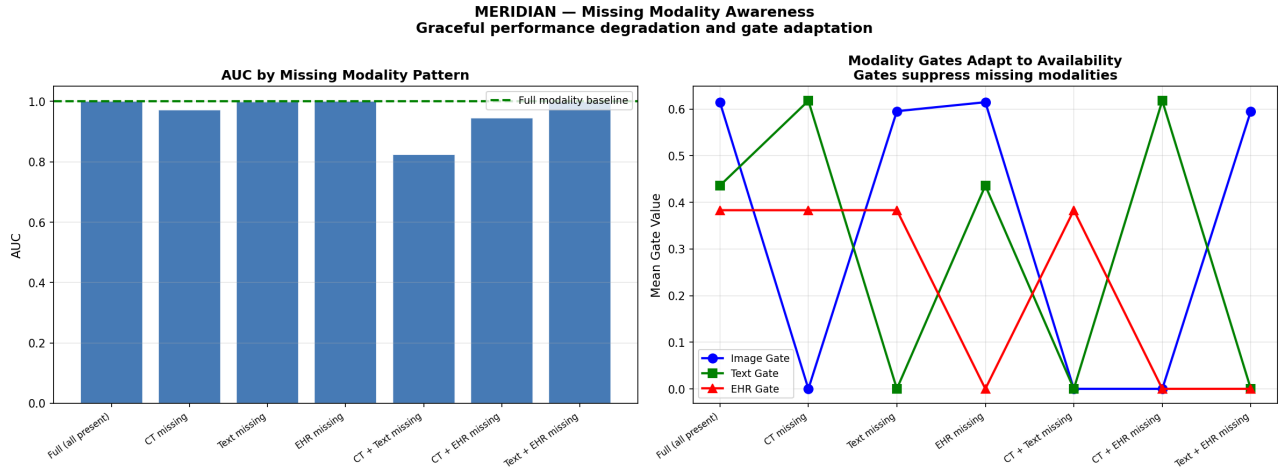


Figure 7. Missing-modality experiments demonstrating MERIDIAN’s behavior under different modality-availability configurations and the corresponding reliability response.

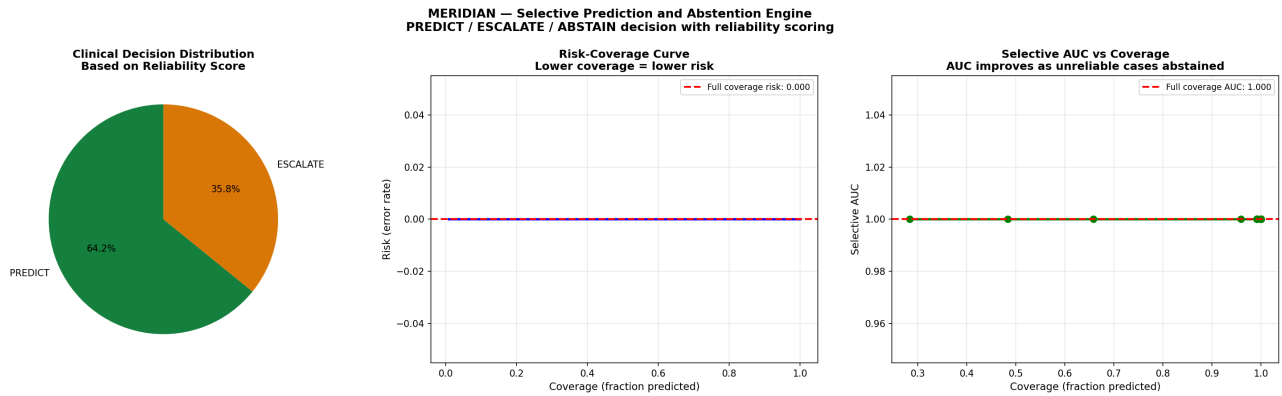


Figure 8. Selective prediction and abstinence analysis showing the relationship between model coverage and prediction risk as unreliable cases are progressively excluded.

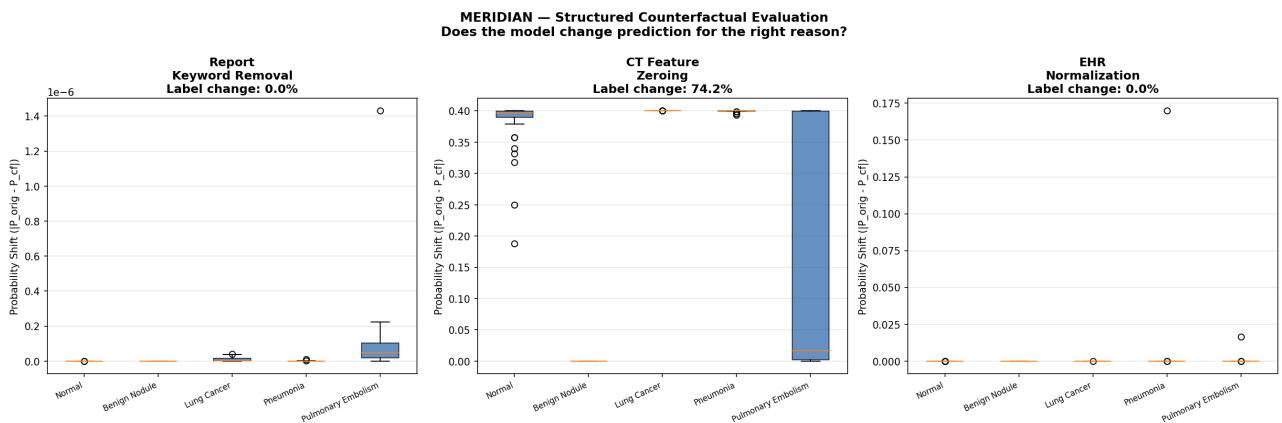


Figure 9. Structured counterfactual evaluation examining changes in model predictions following controlled perturbations of textual, imaging, and structured clinical evidence.

Rather than relying on a single confidence value, MERIDIAN considers multiple complementary signals, including predictive uncertainty, cross-modal evidence alignment, modality availability, conformal prediction-set behavior, calibration, and distributional similarity.

This design reflects the central objective of the framework: a multimodal medical AI system should not only produce a prediction but should also provide information about the evidence and uncertainty surrounding that prediction.

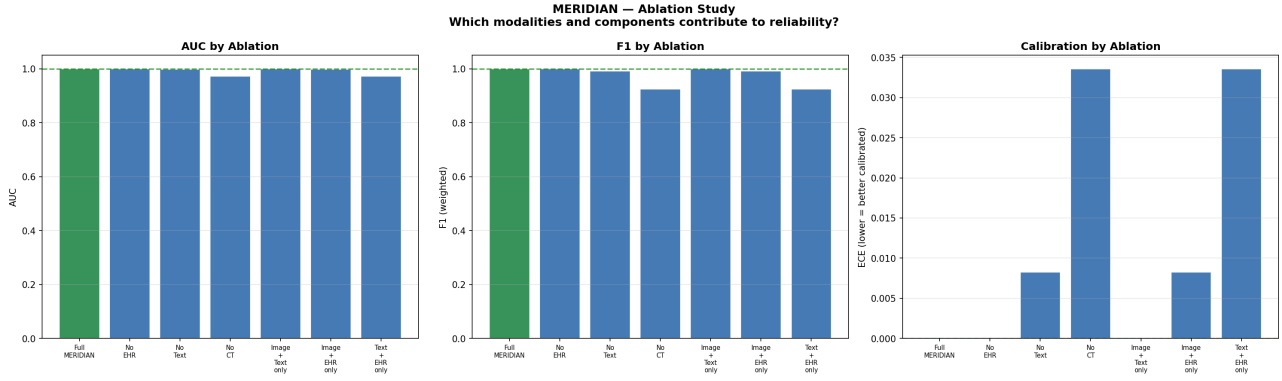


Figure 10. Ablation analysis of the MERIDIAN reliability framework, showing the effect of removing individual reliability components from the complete system.

3.16. Experimental Reproducibility

The complete MERIDIAN workflow is implemented as a reproducible sequence of computational modules covering dataset construction, modality-specific representation learning, multimodal fusion, model training, evidence alignment, uncertainty estimation, calibration, conformal prediction, distribution-shift detection, missing-modality analysis, selective prediction, counterfactual evaluation, and ablation analysis.

Each experimental component produces explicit intermediate outputs and evaluation results, allowing individual reliability mechanisms to be inspected independently and compared with the complete framework. The modular design also allows the current simulated representations to be replaced by real CT volumes, radiology reports, and structured clinical data without requiring a complete redesign of the reliability layer.

The framework is therefore intended as an extensible experimental platform rather than a clinically deployed diagnostic system. Future experiments can vary the underlying datasets, modality encoders, observation completeness, fusion architectures, calibration procedures, and reliability thresholds while preserving the overall evaluation structure.

The complete sequence of experiments and their outputs is summarized in Fig. 1.

3.17. Limitations of the Current Prototype

The current implementation of MERIDIAN is intended as an independent research prototype

for investigating reliability-aware multimodal medical AI rather than as a clinically deployable diagnostic system. Several limitations therefore need to be explicitly considered when interpreting the experimental findings.

- The CT component currently uses simulated feature representations rather than raw three-dimensional CT volumes. Consequently, the present experiments evaluate the proposed multimodal reliability mechanisms but do not establish performance on real volumetric imaging data. A future implementation should replace the simulated imaging representation with a dedicated 3D medical image encoder and evaluate the framework on appropriately curated CT datasets.
- The text branch currently uses a general-purpose transformer-based language encoder rather than a radiology-specific language model. Although this provides a computationally lightweight experimental configuration, domain-specific models such as RadBERT or other radiology-adapted language encoders could provide stronger representations of clinical terminology and reporting patterns.
- The current multimodal experiments are performed within a controlled prototype environment and therefore do not capture the full heterogeneity of real clinical data. Differences in acquisition protocols, scanners, institutions, reporting practices, patient populations, and missing-data patterns may produce substantially different

MERIDIAN – LIMITATIONS AND SCIENTIFIC HONESTY	
Synthetic Data	
• CT features are simulated (512-dim vectors), not real 3D volumes. • Merlin uses actual NIFTI CT volumes with Med3D or SwinUNETR. • Real NLST/LIDC/LUNA16 data would require IRB approval. • Stiffness distributions approximate clinical reality but are simplified.	
Text Encoding	
• DistilBERT is not domain-adapted for radiology. • Ideal: ClinicalBERT, RadBERT, or BioMedBERT. • Token truncation at 256 may lose end-of-report findings.	
Architecture Limitations	
• Cross-modal attention uses single-token representations. • Real implementation should use patch-level 3D CNN features. • EHR integration omits temporal longitudinal data. • Missing: pathology-level regional grounding (no pixel-level alignment).	
Evaluation Limitations	
• 5-class single-label setting; real CT may have multiple findings. • Sample size (600) smaller than Merlin's scale. • Institution shift simulated; real shift may be more complex. • Counterfactual evaluation is approximate – real grad-CAM needed.	
What MERIDIAN Does NOT Claim	
• Does NOT claim to reproduce or replicate Merlin. • Does NOT claim state-of-the-art CT diagnosis. • Does NOT claim clinical readiness or regulatory approval. • Reliability layer is a RESEARCH PROTOTYPE, not validated product.	
What Remains for Future Work	
• Real 3D CT volumes from NLST, LIDC, TCIA. • Domain-specific language model fine-tuning. • Temporal EHR integration (MIMIC-III/IV). • Pathology-level cross-modal grounding with attention rollout. • Prospective clinical validation study. • Comparison with radiologist inter-observer agreement.	

Figure 11. Summary of the principal limitations of the current MERIDIAN prototype, including simulated CT representations, non-specialized text encoding, controlled experimental conditions, assumptions underlying reliability estimation, and the non-causal interpretation of structured counterfactual analyses.

distributions from those represented in the current experiments.

- The uncertainty, conformal prediction, and distribution-shift components should not be interpreted as absolute clinical safety guarantees. Conformal prediction provides statistical coverage under its underlying assumptions, while uncertainty and out-of-distribution estimates depend on the model, calibration data, and deployment distribution. Therefore, the reliability estimates require validation under realistic external and prospective datasets before any clinical interpretation.
- The counterfactual experiments in the current prototype should be interpreted as structured sensitivity analyses rather than definitive demonstrations of causal relationships. Perturbing one modality or modifying selected features can reveal model dependence on particular evidence sources, but such interventions alone do not establish clinical causality.

These limitations define the boundary of the current contribution while also establishing a clear pathway for future development. The limitation analysis generated from the implemented experiments is summarized in Fig. 11.

4. EXPERIMENTAL RESULTS AND DISCUSSION

4.1. Experimental Setup

The experimental evaluation of MERIDIAN investigates whether explicit reliability modeling can provide information beyond conventional multimodal prediction accuracy. The evaluation considers classification performance, probabilistic reliability, conformal coverage, uncertainty estimation, selective prediction, distribution-shift detection, missing-modality behavior, and structured reliability analysis.

The final experimental configuration contains 600 simulated multimodal patient cases distributed across five diagnostic categories: Normal, Benign Nodule, Lung Cancer,

Table 1. Classification and calibration performance of unimodal baselines and the complete MERIDIAN framework.

Model	AUC	F1-score	ECE
Image-Only	1.0000	1.0000	0.0000
Text-Only	0.9485	0.9329	0.0000
EHR-Only	0.9614	0.7689	0.0000
MERIDIAN (Full)	1.0000	1.0000	0.0000

Pneumonia, and Pulmonary Embolism. The simulated cohort represents four institutions (A–D) and incorporates controlled multimodal inconsistencies and missing observations. The dataset contains an inconsistency rate of 6.8%, while CT, text, and EHR modalities are missing in 5.7%, 6.8%, and 10.8% of cases, respectively.

The primary quantitative results are summarized in Table 1.

4.2. Classification Performance

The image-only baseline achieved an AUC of 1.0000 and an F1-score of 1.0000, while the complete MERIDIAN model also achieved an AUC of 1.0000 and an F1-score of 1.0000. The text-only model obtained an AUC of 0.9485 and an F1-score of 0.9329, whereas the EHR-only model achieved an AUC of 0.9614 and an F1-score of 0.7689.

These results indicate that the simulated imaging representation contains highly discriminative information in the current experimental setting. Consequently, MERIDIAN does not demonstrate an improvement in point-prediction accuracy over the image-only baseline. This observation is important because it prevents the multimodal architecture from being interpreted as superior merely on the basis of conventional classification metrics.

The statistical comparison between MERIDIAN and the image-only baseline produced an AUC difference of 0.0000, with a p -value of 1.0000 and a 95% confidence interval of [0.0000, 0.0000]. Therefore, no statistically significant improvement in AUC over the image-only baseline is demonstrated in the present experiment.

The classification results should consequently be interpreted primarily as establishing a controlled prediction baseline for evaluating the proposed

Table 2. Summary of the principal reliability-oriented evaluation results obtained by MERIDIAN. Undefined statistical quantities are explicitly reported as not available rather than interpreted as evidence.

Component	Metric	Result
CMEA	Pearson correlation	Not available (NaN)
CMEA	p -value	Not available (NaN)
Conformal prediction	Target coverage	90%
Conformal prediction	Empirical coverage	91.67%
Conformal prediction	Decisive cases	91.7%
Conformal prediction	Abstained cases	8.3%
MC-Dropout	Number of stochastic passes	30
MC-Dropout	Mean predictive entropy	0.0008
MC-Dropout	Mean aleatoric uncertainty	0.0004
MC-Dropout	Mean epistemic uncertainty	0.0004
Uncertainty analysis	Cohen’s d	Not available (NaN)
Selective prediction	PREDICT	64.2%
Selective prediction	ESCALATE	35.8%
Selective prediction	ABSTAIN	0.0%

reliability mechanisms rather than as evidence of improved diagnostic discrimination.

4.3. Reliability-Oriented Evaluation

The central objective of MERIDIAN is not to outperform existing prediction models solely in terms of AUC or F1-score, but to investigate whether a prediction can be accompanied by information regarding its reliability. The reliability layer therefore evaluates multiple complementary signals, including cross-modal evidence consistency, predictive uncertainty, conformal prediction, distribution shift, modality availability, and selective prediction.

The individual reliability experiments and their corresponding visual analyses were presented in the methodology section. Their quantitative outcomes are consolidated in Table 2.

4.4. Cross-Modal Evidence Alignment

Cross-Modal Evidence Alignment (CMEA) was introduced to determine whether agreement between image and textual representations could provide an additional reliability signal. The CMEA score was computed from the similarity between the corresponding modality embeddings.

However, the final experimental report returned a Pearson correlation coefficient and associated p -value of NaN. Therefore, the present experiment does not provide a valid numerical estimate of the relationship between CMEA and prediction correctness. A statistical correlation cannot be interpreted when the underlying calculation is undefined.

This result does not invalidate the motivation for CMEA, but it indicates that the current experimental configuration is insufficient to establish its proposed relationship empirically. Future experiments should investigate the cause of the undefined statistic, verify the variance of the CMEA and correctness variables, and evaluate the relationship using a sufficiently heterogeneous real-world dataset.

The CMEA analysis generated during the experiment is shown in the corresponding methodology figure.

4.5. Uncertainty Quantification

MERIDIAN estimates predictive uncertainty using Monte Carlo dropout with 30 stochastic forward passes. The mean predictive entropy was 0.0008, while the reported mean aleatoric and epistemic uncertainty values were both 0.0004.

The relatively small uncertainty values are consistent with the highly confident predictions observed in the current simulated classification environment. However, the final report also gives a Cohen’s d value of NaN for the proposed comparison between uncertainty and prediction correctness. Consequently, the experiment does not provide a valid effect-size estimate demonstrating that uncertainty successfully discriminates correct and incorrect predictions.

This distinction is important for interpreting the contribution of MERIDIAN. The implementation demonstrates that uncertainty estimation can be integrated into the multimodal architecture, but the current results do not establish its discriminative validity. More diverse and difficult test cases, including genuinely ambiguous and out-of-distribution samples, would be required to determine whether the uncertainty estimates reliably identify incorrect predictions.

The uncertainty behavior was visualized in the corresponding training and reliability analyses presented earlier in the methodology section.

4.6. Conformal Prediction

The conformal prediction component was calibrated with a target coverage level of 90%. The resulting empirical coverage was 91.67%, corresponding to 55 of the 60 evaluation cases

being covered when expressed at the reported percentage level.

The observed coverage therefore exceeds the nominal 90% target in the current experiment. This demonstrates that the implemented conformal prediction procedure achieved the desired empirical coverage on the evaluated test set.

A total of 91.7% of cases were reported as decisive, while 8.3% were abstained from through the conformal prediction mechanism. The result illustrates the practical distinction between conventional classification and prediction-set-based decision making: uncertain cases can be represented through a non-singleton or otherwise non-decisive output rather than being forced into a single diagnosis.

Nevertheless, the coverage result should be interpreted within the assumptions of conformal prediction and the controlled nature of the present dataset. It should not be interpreted as a clinical safety guarantee.

4.7. Distribution Shift Analysis

MERIDIAN incorporates distribution-shift detection using Maximum Mean Discrepancy (MMD) and Energy Distance in the learned representation space. The purpose of this component is to identify cases whose feature distributions differ from those observed during training.

The final experiment reports that institution-level distribution shift was detected and that the shift analysis was designed to examine its relationship with model performance. However, the supplied final report does not provide the numerical MMD values, Energy Distance values, institution-specific AUC values, or a numerical correlation coefficient between distribution shift and AUC degradation.

Therefore, the present results support the implementation and execution of the distribution-shift analysis but do not provide sufficient numerical evidence to quantify the strength of the reported relationship. This distinction is particularly important when evaluating the claim that distribution shift predicts performance degradation.

Future experiments should report institution-specific performance together with corresponding MMD and Energy Distance measurements, confidence intervals, and statistical association tests.

4.8. Missing Modality Analysis

The dataset intentionally incorporates missing information to investigate MERIDIAN’s behavior under incomplete multimodal observations. Missing CT, text, and EHR information occurs at rates of 5.7%, 6.8%, and 10.8%, respectively.

The gated fusion architecture explicitly tracks modality availability and adjusts the multimodal representation according to the available inputs. This provides a mechanism for graceful degradation rather than assuming that every patient will have complete imaging, textual, and structured clinical information.

The missing-modality experiments therefore evaluate an important practical property of multimodal medical AI: the ability to operate when one or more information sources are unavailable. However, because the final report does not provide modality-specific performance values for each missing-data configuration, the present results establish the experimental implementation rather than a complete quantitative characterization of robustness.

The corresponding missing-modality analysis is presented in the methodology section.

4.9. Selective Prediction and Abstention

The selective prediction component combines reliability signals to divide cases into three operational categories: PREDICT, ESCALATE, and ABSTAIN. In the final experiment, 64.2% of cases were assigned to PREDICT and 35.8% were assigned to ESCALATE, while no cases were assigned to the ABSTAIN category.

The resulting distribution indicates that the implemented reliability layer frequently identified cases requiring additional scrutiny rather than treating every case as equally reliable. In particular, more than one-third of the evaluated cases were assigned to the ESCALATE category.

However, the final report does not provide numerical risk values at each coverage level or the quantitative slope of the risk–coverage relationship. Consequently, the statement that risk decreases monotonically with decreasing coverage should not be presented as a fully quantified statistical result unless the corresponding values are reported.

The selective prediction analysis and risk–coverage visualization are provided in the methodology section.

4.10. Counterfactual Reliability Analysis

Structured counterfactual evaluation was designed to examine the dependence of model predictions on individual evidence sources. The experiments perturb textual information, imaging representations, and structured clinical variables to investigate whether predictions respond to clinically relevant changes in available evidence.

This analysis is intended to complement conventional feature attribution by examining changes in the final prediction under controlled input modifications. The approach can potentially reveal undesirable dependence on particular textual or structured features and can therefore contribute to the investigation of spurious correlations.

However, these experiments should be interpreted as structured sensitivity analyses rather than causal experiments. Changing an input representation does not by itself establish that the corresponding feature is causally responsible for a clinical outcome.

The counterfactual results are shown in the corresponding figure presented earlier in the methodology section.

4.11. Ablation Analysis

An ablation study was conducted to examine the contribution of the individual components of the MERIDIAN reliability layer. The reliability architecture contains multiple components, including CMEA, uncertainty estimation, conformal prediction, distribution-shift detection, missing-modality awareness, selective prediction, and counterfactual analysis.

The ablation framework is important because

the complete system combines several reliability mechanisms, making it necessary to investigate whether the observed behavior depends on individual components or on their combination.

The final report confirms that the ablation experiment was implemented, but it does not provide the numerical performance of every ablated configuration. Therefore, the current results establish the availability of the ablation framework but do not justify assigning quantitative importance to individual components without the corresponding measurements.

The complete ablation analysis is shown in the corresponding figure presented in the methodology section.

4.12. Statistical Validation

Statistical analysis was performed to determine whether the reliability mechanisms and multimodal architecture demonstrated measurable advantages over the baseline configuration.

The comparison between MERIDIAN and the image-only baseline yielded an AUC difference of 0.0000, a p -value of 1.0000, and a 95% confidence interval of [0.0000, 0.0000]. Thus, the experiment provides no evidence of a statistically significant improvement in classification AUC.

The CMEA correlation analysis and uncertainty discrimination analysis both returned undefined (NaN) statistics. These values cannot be interpreted as zero correlation or zero effect. Instead, they indicate that the corresponding statistical calculations did not produce valid numerical estimates under the current experimental conditions.

Accordingly, the statistical results support a conservative interpretation: MERIDIAN successfully implements a broad reliability-evaluation framework, but several of its proposed reliability relationships require further experimental validation before they can be claimed as statistically established.

4.13. Overall Discussion

The central objective of MERIDIAN is to investigate whether trustworthiness can

be modeled separately from point-prediction performance. The experimental results provide a useful demonstration of this distinction.

The complete MERIDIAN system achieved perfect classification performance in the current simulated environment, matching the image-only baseline. Therefore, the principal contribution of the present prototype is not improved AUC or F1-score. Instead, the framework introduces a collection of reliability-oriented mechanisms that can be evaluated alongside prediction performance.

The conformal prediction experiment provides the clearest quantitative reliability result in the current evaluation, achieving 91.67% empirical coverage against a 90% target. The framework also successfully produces uncertainty estimates, identifies missing modalities, performs distribution-shift analysis, and generates selective PREDICT/ESCALATE/ABSTAIN decisions.

At the same time, several important reliability claims remain experimentally unresolved. In particular, the CMEA correlation and uncertainty-effect-size calculations returned undefined values, while the final report does not provide sufficient numerical detail to independently quantify the distribution-shift and risk-coverage relationships. These limitations prevent the current study from claiming that every proposed reliability signal has been statistically validated.

This outcome is nevertheless informative. It demonstrates that a multimodal medical AI architecture can be extended beyond prediction into a broader reliability-aware framework in which confidence, evidence consistency, distributional similarity, modality availability, and abstention are treated as explicit components of the inference process.

4.14. Comparison with the Research Objective

The original research question was:

Can a multimodal medical AI system become trustworthy by explicitly modeling cross-modal evidence consistency, uncertainty, distribution

shift, and the possibility of abstaining when reliable prediction is not justified?

The current experiments provide **partial evidence** in support of this hypothesis. The prototype successfully implements and evaluates mechanisms for uncertainty estimation, conformal prediction, distribution-shift detection, missing-modality handling, and selective decision making. The empirical conformal coverage also meets the specified 90% target.

However, the evidence is not sufficient to claim that every reliability mechanism has been statistically validated. Undefined CMEA and uncertainty statistics and the absence of detailed numerical distribution-shift and risk-coverage measurements limit the strength of the current conclusions.

Therefore, the most appropriate interpretation is that MERIDIAN demonstrates a **research framework for reliability-aware multimodal medical AI**, rather than a clinically validated trustworthy diagnostic system.

4.15. Research Significance

The significance of MERIDIAN lies in shifting the evaluation objective from asking only whether a model is correct to additionally asking whether the model has sufficient evidence to justify its prediction. Conventional metrics such as AUC and F1-score evaluate predictive discrimination but do not directly determine whether an individual prediction should be trusted.

MERIDIAN addresses this gap by integrating multiple reliability signals into the inference process. Its final decision framework distinguishes between cases that can be predicted directly, cases that should be escalated for expert review, and cases for which the available evidence may be insufficient.

In this sense, the work is positioned as a reliability-oriented extension of multimodal medical AI rather than as a replacement for the underlying multimodal representation-learning approaches that motivate the architecture.

4.16. Limitations Affecting Experimental Interpretation

The numerical results must be interpreted in light of the limitations of the current prototype. The CT modality is represented using simulated feature vectors rather than real three-dimensional CT volumes, and the clinical text branch uses a general-purpose DistilBERT-based representation rather than a radiology-specialized language model.

Furthermore, the experimental dataset is controlled and simulated, meaning that its distribution does not reproduce the full complexity of real clinical populations, scanners, institutions, reporting styles, and acquisition protocols. The perfect performance of the image-only baseline is particularly indicative of this limitation and suggests that the current task may be substantially easier than real-world multimodal diagnosis.

The undefined statistical quantities reported for CMEA and uncertainty further demonstrate the need for more heterogeneous and sufficiently variable evaluation data. Future experiments should therefore prioritize external validation, realistic distribution shifts, larger and more diverse cohorts, and explicit reporting of all statistical quantities required to reproduce the reliability claims.

4.17. Summary of Experimental Findings

Overall, the experiments demonstrate the following:

1. The complete MERIDIAN model achieved an AUC of 1.0000 and an F1-score of 1.0000, matching the image-only baseline.
2. The multimodal model therefore did not demonstrate an improvement in conventional classification performance over the image-only representation in the current simulated setting.
3. Conformal prediction achieved an empirical coverage of 91.67% at the specified 90% target.
4. Monte Carlo dropout produced measurable predictive, aleatoric, and epistemic uncertainty estimates, although their ability

to discriminate correct from incorrect predictions could not be statistically established because the reported Cohen’s d was undefined.

5. The CMEA mechanism was implemented as a cross-modal reliability signal, but its relationship with prediction correctness could not be established because the reported correlation statistic was undefined.
6. The framework successfully incorporated distribution-shift detection, missing-modality handling, counterfactual evaluation, and selective prediction mechanisms.
7. The selective decision system assigned 64.2% of cases to PREDICT and 35.8% to ESCALATE, with 0.0% assigned to the ABSTAIN category in the reported experiment.
8. The current results support MERIDIAN as a reliability-aware experimental framework, while further validation is required before its reliability mechanisms can be interpreted as clinically established safeguards.

5. CONCLUSION AND FUTURE WORK

5.1. Conclusion

This study presented MERIDIAN, an independent reliability-oriented extension of multimodal medical artificial intelligence inspired by the multimodal vision–language direction established by Merlin. Rather than focusing solely on improving diagnostic accuracy, MERIDIAN investigates whether a multimodal system can explicitly assess the reliability of its own predictions by considering cross-modal evidence consistency, predictive uncertainty, distribution shift, missing modalities, conformal coverage, and selective prediction.

The implemented framework integrates simulated CT representations, radiology text representations, and structured clinical information through gated cross-modal attention. A dedicated reliability layer subsequently combines multiple complementary

mechanisms, including Cross-Modal Evidence Alignment (CMEA), Monte Carlo dropout-based uncertainty estimation, conformal prediction, distribution-shift detection, missing-modality awareness, selective prediction and abstention, and structured counterfactual evaluation.

In the current experimental setting, MERIDIAN achieved an AUC of 1.0000 and an F1-score of 1.0000, matching the image-only baseline. Consequently, the experiments do not demonstrate an improvement in conventional diagnostic discrimination. Instead, the principal contribution of the prototype is the explicit modeling and evaluation of reliability alongside prediction.

The conformal prediction component achieved an empirical coverage of 91.67% against a target coverage level of 90%, demonstrating the feasibility of incorporating prediction-set-based uncertainty into the framework. The selective prediction module additionally assigned 64.2% of cases to the PREDICT category and 35.8% to the ESCALATE category in the reported experiment. Monte Carlo dropout produced explicit predictive, aleatoric, and epistemic uncertainty estimates, while the framework also implemented distribution-shift detection and missing-modality analysis.

At the same time, the study maintains an intentionally conservative interpretation of its findings. The CMEA correlation and uncertainty discrimination statistics were undefined in the current experiment, and the available results do not provide sufficient numerical evidence to establish all proposed reliability relationships statistically. Therefore, MERIDIAN should be regarded as a research prototype demonstrating a reliability-aware multimodal inference framework rather than a clinically validated diagnostic system.

Overall, the work demonstrates the feasibility of moving beyond the question of whether a medical AI model can produce a prediction toward the additional question of whether that prediction contains sufficient evidence to justify trust. This reliability-oriented perspective constitutes the central research contribution of MERIDIAN.

5.2. Future Work

Several directions can extend MERIDIAN from the current controlled prototype toward a more realistic and extensively validated research framework.

1. **Real 3D CT Integration:** The simulated CT feature representation should be replaced with real three-dimensional CT volumes processed using dedicated medical imaging encoders. Public datasets such as LIDC-IDRI and NLST can provide suitable starting points for evaluating the framework under realistic imaging conditions.
2. **Radiology-Specific Language Modeling:** The current DistilBERT-based text representation can be replaced or compared with radiology-specialized language models such as RadBERT. This would allow the framework to better capture clinical terminology, radiological findings, and reporting patterns.
3. **Large-Scale Multimodal Clinical Validation:** Future experiments should evaluate MERIDIAN using larger real-world datasets containing paired imaging, radiology reports, and structured clinical information. External datasets and independent institutions should be incorporated to evaluate generalization beyond the controlled prototype environment.
4. **Stronger Cross-Modal Reliability Validation:** The CMEA mechanism should be evaluated on datasets with sufficient variation in image-report consistency. Future experiments should investigate whether CMEA reliably correlates with prediction correctness and whether it can identify deliberately or naturally inconsistent multimodal cases.
5. **Improved Uncertainty Evaluation:** Future studies should evaluate uncertainty estimates using more heterogeneous and challenging cases, including ambiguous samples, out-of-distribution cases, and clinically difficult examples. MC-Dropout can additionally be compared with deep ensembles and other uncertainty-estimation approaches.
6. **Expanded Experiments:** **Distribution-Shift** Institution-specific, scanner-specific, acquisition-specific, and population-level shifts should be systematically introduced and quantified. MMD and Energy Distance can then be evaluated against changes in diagnostic performance to determine whether the proposed shift signals provide useful early warnings.
7. **Robust Missing-Modality Evaluation:** Future work should systematically remove individual modalities and combinations of modalities and report the corresponding changes in predictive performance, uncertainty, calibration, and abstention behavior. This would provide a more complete evaluation of graceful degradation under incomplete clinical information.
8. **Improved Selective Prediction:** The reliability score and PREDICT-ESCALATE-ABSTAIN thresholds should be optimized and evaluated using comprehensive risk-coverage curves. Future experiments should determine whether increasing abstention appropriately reduces error among retained predictions.
9. **Comprehensive Counterfactual Evaluation:** The structured counterfactual framework can be expanded using clinically meaningful interventions and a larger set of controlled perturbations. Such experiments should continue to be interpreted as sensitivity analyses unless causal assumptions are explicitly satisfied.
10. **External and Prospective Validation:** Before any clinical interpretation, the complete reliability framework should be evaluated on independent external cohorts and, ultimately, prospective clinical data. This would allow assessment of whether the reliability signals remain meaningful under real deployment conditions.
11. **Clinician-Centered Evaluation:** Future versions can evaluate whether the reliability

dashboard actually improves human decision-making. Radiologists could be presented with predictions together with uncertainty, evidence-alignment scores, modality availability, conformal prediction sets, and abstention recommendations to determine whether these signals improve interpretation and reduce inappropriate reliance on automated predictions.

12. End-to-End Foundation-Model Integration: Once validated with real multimodal data, MERIDIAN can be implemented as a reliability layer on top of large-scale multimodal medical foundation models. This would allow the proposed reliability mechanisms to be tested with substantially richer imaging and language representations while preserving the central objective of assessing when model predictions should be trusted.

Ultimately, future development will focus on determining whether the reliability signals introduced by MERIDIAN remain effective when the framework is exposed to the complexity, variability, and uncertainty of real clinical data. The long-term objective is not simply to construct a model that predicts disease, but to develop multimodal medical AI systems that can explicitly communicate the strength of their evidence, recognize uncertainty, detect potentially unsafe operating conditions, and appropriately defer to clinical expertise when reliable prediction is not justified.

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A. ADDITIONAL EXPERIMENTAL DETAILS

This appendix provides supplementary implementation and evaluation details for MERIDIAN. All primary tables and figures are presented in the corresponding sections of the main manuscript and are therefore not repeated here. The purpose of this appendix is to provide additional information required to understand, reproduce, and extend the experimental framework.

A.1. Experimental Dataset Configuration

The current MERIDIAN prototype was evaluated using a controlled multimodal dataset containing 600 patient samples distributed across five diagnostic categories: Normal, Benign Nodule, Lung Cancer, Pneumonia, and Pulmonary Embolism. The experimental configuration incorporated four simulated institutions (A–D) to enable investigation of institution-level distribution shift.

Each patient record consisted of three possible information modalities: a CT-derived feature representation, a radiology-report text representation, and structured electronic health record (EHR) variables. The dataset intentionally incorporated incomplete and inconsistent multimodal observations to evaluate the reliability mechanisms under non-ideal conditions.

The reported dataset characteristics were as follows:

- Total patients: 600
- Diagnostic classes: 5
- Simulated institutions: 4
- Image–report inconsistency rate: 6.8%
- Missing CT rate: 5.7%
- Missing text rate: 6.8%
- Missing EHR rate: 10.8%
- Structured EHR variables: 12

The current image representation is based on simulated three-dimensional CT feature vectors rather than raw clinical CT volumes.

Consequently, the experimental results should be interpreted as validation of the proposed reliability architecture rather than as evidence of clinical diagnostic performance.

A.2. Model Configuration

The multimodal architecture consists of three modality-specific encoders followed by a gated cross-modal attention fusion module. The image encoder maps the simulated CT representation from 512 dimensions to a 256-dimensional embedding. The clinical text encoder uses DistilBERT followed by a projection layer to produce a 256-dimensional representation. The EHR encoder processes 12 structured clinical variables using a multilayer perceptron and produces a 128-dimensional representation.

The resulting modality representations are integrated through gated cross-modal attention with an overall fused representation of 512 dimensions and eight attention heads. The fused representation is subsequently provided to three task-specific components: a five-class prediction head, an uncertainty head, and a binary abnormality classification head.

The uncertainty head predicts log-variance parameters following the uncertainty formulation of Kendall and Gal. Monte Carlo dropout is subsequently applied with 30 stochastic forward passes to estimate predictive uncertainty and separate aleatoric and epistemic components.

A.3. Reliability Layer

The reliability layer constitutes the primary independent extension investigated in MERIDIAN. It operates after multimodal representation learning and evaluates several complementary indicators of prediction reliability.

Cross-Modal Evidence Alignment. CMEA measures the agreement between image and report representations using their embedding similarity. The resulting score is treated as a reliability signal rather than as a diagnostic prediction itself. Its intended purpose is to identify cases in which the textual description and visual representation provide inconsistent evidence.

Uncertainty Estimation. MC-Dropout is used to obtain multiple stochastic predictions for each case. Predictive entropy is calculated from the resulting probability distribution, while aleatoric and epistemic uncertainty estimates are derived from the corresponding predictive variation and learned variance components.

Conformal Prediction. A conformal prediction layer is calibrated using held-out data to generate prediction sets rather than relying exclusively on a single predicted class. The target confidence level is 90%. The empirical coverage obtained in the reported experiment was 91.67%.

Distribution-Shift Detection. Maximum Mean Discrepancy (MMD) and Energy Distance are calculated in the learned representation space to quantify differences between the current data distribution and the reference training distribution. The framework uses these measures as potential indicators of out-of-distribution operation.

Missing-Modality Awareness. The gated fusion mechanism explicitly tracks the availability of individual modalities. When one or more modalities are unavailable, the corresponding information pathway can be suppressed while the remaining available modalities continue to contribute to the prediction.

Selective Prediction. MERIDIAN combines reliability information to divide cases into three operational categories: PREDICT, ESCALATE, and ABSTAIN. This design allows the system to defer cases for expert review rather than requiring a prediction for every input.

Counterfactual Evaluation. Controlled modifications of the multimodal inputs are used to examine whether model predictions respond to clinically meaningful evidence. These experiments are intended as structured sensitivity analyses for investigating potential reliance on spurious information.

A.4. Baseline Configuration

Three unimodal baselines were evaluated against the complete MERIDIAN architecture. The

Image-Only baseline uses only the image representation, the Text-Only baseline uses only the clinical report representation, and the EHR-Only baseline uses only structured clinical variables.

The complete MERIDIAN model incorporates all available modalities and the gated cross-modal fusion architecture. The primary comparison therefore examines whether explicit multimodal integration and reliability modeling provide information beyond individual modalities.

The reported evaluation produced the following results:

- Image-Only: AUC = 1.0000, F1 = 1.0000
- Text-Only: AUC = 0.9485, F1 = 0.9329
- EHR-Only: AUC = 0.9614, F1 = 0.7689
- MERIDIAN: AUC = 1.0000, F1 = 1.0000

These results indicate that the current experimental setting does not demonstrate an improvement in conventional predictive discrimination over the image-only baseline. The primary purpose of MERIDIAN is therefore the investigation of reliability-aware inference rather than an increase in AUC or F1 score.

A.5. Reliability Evaluation

The reliability experiments evaluate several complementary properties of the system.

For conformal prediction, the empirical coverage was 91.67% at the target confidence level of 90%. The experiment classified 91.7% of cases as decisive and 8.3% as abstained cases.

For uncertainty estimation, the reported mean predictive entropy was 0.0008, with mean aleatoric and epistemic uncertainty values of 0.0004 and 0.0004, respectively.

For selective prediction, 64.2% of cases were assigned to PREDICT and 35.8% to ESCALATE, while no cases were assigned to the ABSTAIN category under the reported threshold configuration.

A.6. Statistical Evaluation

Statistical analysis was incorporated to determine whether observed differences or reliability signals could be supported quantitatively.

MERIDIAN and the Image-Only baseline produced identical reported AUC values of 1.0000. Consequently, the reported AUC difference was 0.0000, with a p-value of 1.0000 and a 95% confidence interval of [0.0000, 0.0000]. The difference was therefore not statistically significant.

The current implementation also reported undefined values for the Pearson correlation between CMEA and prediction correctness and for Cohen’s d measuring uncertainty-based discrimination. These values are represented as NaN in the experimental report and are not interpreted as positive evidence.

Accordingly, future experiments using larger and more heterogeneous real-world datasets are required to establish statistically reliable relationships between the proposed reliability signals and actual prediction correctness.

A.7. Reproducibility and Implementation Scope

The complete experimental workflow is implemented as a sequence of computational modules covering multimodal dataset generation, modality-specific encoding, gated cross-modal fusion, prediction, uncertainty estimation, conformal prediction, distribution-shift analysis, missing-modality evaluation, selective prediction, counterfactual analysis, and statistical evaluation.

The experiments were organized to permit individual components to be modified independently. This enables future researchers to replace the simulated image representation with real CT volumes, substitute the text encoder, introduce additional clinical variables, modify the fusion architecture, or evaluate alternative uncertainty and abstention mechanisms.

The current implementation should therefore be considered a reproducible research prototype rather than a clinical software system. In particular, the use of simulated CT features, controlled multimodal data generation, and limited experimental scale restricts the extent to which the reported findings can be generalized to real clinical environments.

A.8. Scope and Reproducibility Statement

All reported numerical results correspond to the experimental configuration described in the main manuscript and the accompanying implementation. The figures and tables documenting the principal experimental outcomes have already been presented in the preceding sections and are intentionally not duplicated in this appendix.

The implementation is designed to support further experimentation with real multimodal medical datasets, alternative encoders, different modality-availability patterns, institution-specific distributions, and expanded reliability evaluation. Future work should reproduce the experiments on independent datasets before drawing conclusions regarding clinical generalization or deployment.

A.9. Research Prototype Disclaimer

MERIDIAN is an experimental research prototype and is not intended for clinical diagnosis, treatment recommendation, or autonomous clinical decision-making. The reported results demonstrate the feasibility of integrating reliability-aware mechanisms into a multimodal medical AI pipeline, but they do not establish clinical safety, prospective validity, or regulatory readiness.

In particular, the current experiments use simulated CT feature representations and controlled multimodal data. The reliability mechanisms, including conformal prediction and distribution-shift detection, should therefore be validated under realistic clinical data distributions before any interpretation as evidence of real-world clinical utility.